

Land Survey

## Engineering Mathematics III

### Assignment 1 - Questions Only

Instructions: Answer all questions. Show all working and use correct units.

#### Section A

1. State the meaning of the formula below and identify its main symbols. **(4 marks)**

$$\bar{x} = \frac{\sum x}{n}$$

2. State the meaning of the formula below and identify its main symbols. **(4 marks)**

$$s = \sqrt{\frac{\sum (x - \bar{x})^2}{n - 1}}$$

3. State the meaning of the formula below and identify its main symbols. **(4 marks)**

$$P(A) = \frac{n(A)}{n(S)}$$

4. State the meaning of the formula below and identify its main symbols. **(4 marks)**

$$AX = B$$

#### Section B

5. Calculate mean and standard deviation of repeated observations. **(10 marks)**
6. Solve a matrix system from survey adjustment. **(10 marks)**
7. Compute probability from survey quality data. **(10 marks)**
8. Create one realistic land survey problem based on this topic and solve it fully. **(20 marks)**